

Challenges with Modern Data Centers, Design Considerations and Recommended Power System Studies

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Abstract — Data Center power systems today face unique power demand challenges due to the proliferation of cloud computing, Artificial Intelligence (AI), Internet of Things (IOT), as well as the intensifying demand for high bandwidth applications which cause instantaneous spikes in load fluctuations. In addition, resonance incidents observed recently in new data centers are leading to power quality concerns. Motivated by such challenges, this paper discusses the design considerations for modern data centers and suggests various power system studies during engineering design to overcome such challenges. This paper presents a literature review that suggests the essential power system studies that can help identify the root cause of the power quality concerns, identify potential points of failure during operations of an existing data center power system, and allow designers and engineers to overcoming such challenges when designing modern data center power systems and developing and refining equipment specifications.

Keywords— Artificial Intelligence, Coefficient of Grounding, Data Center power system, Graphics Processing Unit, Internet of Things, Information Technology Load, Power Electronics Non-linear Load, Point of Common Coupling, Sub-resonance, Power System Studies, Power Transients, Transient Stability.

I INTRODUCTION

A. Background

Fundamental drivers of modern data centers include the increasing necessity for data storage and computational capability, predominantly driven by the proliferation of cloud computing, AI, IoT, as well as the intensifying demand for high-bandwidth applications, necessitating a resilient infrastructure to effectively manage the substantial quantities of data produced by users and businesses. Historically, data centers were physical spaces engineered to support an organization's requirements; however, they have evolved into essential frameworks that support the digital economy, especially in relation to AI and cloud computing. [1] defines data centers as facilities that host server systems, computer systems, and associated components such as cooling units, redundancy power supplies, and power storage systems,

serving as critical infrastructure for managing and processing large amounts of data. They serve as critical infrastructure in the digital landscape, enabling organizations and individuals to access online services and applications efficiently. [4] defines data centers as entities that include physical servers, virtual servers, operating systems, applications, middleware, and application services. They can be virtual or physical and represent the highest level of resources offered by a cloud service provider. While the definition of a data center may somewhat vary across literature, this paper adopts the definition provided in [5] which describes a data center as a facility that constitutes a specialized infrastructure that centralizes IT operations and apparatus for data and application storage, processing, and management. A modern data center may consume over 100 MW electric power and uses an advanced power distribution system to ensure reliability and power quality [6]. The main components of a data center include the electrical distribution equipment, cooling systems, servers, storage devices, networking and communication equipment, software, security systems and data center staff [7] [8]. Data Centers are categorized by the Uptime Institute on a scale from tier 1 (least sophisticated, fundamental infrastructure with minimal redundancy) to tier 4 (most sophisticated, exhibiting complete fault tolerance with redundancy integrated across all components), where each tier delineates progressive levels of reliability and operational continuity predicated upon power and cooling pathways, maintenance capabilities, and redundancy attributes [10].

By 2030, global data traffic is forecasted to reach approximately 5016 exabytes (EB) or approximately 5 zettabytes (ZB), with an annual increase of 55% from 2020 [11]. This growth reflects the rising demand for data transfer and digital services in future wireless networks. Globally data centers are growing rapidly and significantly to handle the rapidly growing data traffic, while the energy demand of the data centers is also increasing. The expected energy consumption of data centers by 2030 is projected to exceed 1 Peta-Watt hour (PWh) annually, driven by the exponential growth of data traffic and cloud computing technologies [12]. Data centers are anticipated to account for 4.5% of global energy consumption by 2025, with further increases expected by 2030 [13]. Electric Power Research Institute (EPRI)

estimates that data centers in the USA could grow to consume up to 9% of its electricity generation annually by 2030, up from 4% of the total load in 2023.

B. Challenges

In a data center, more so in a co-location setting where the data center infrastructure is shared by several users, users continuously adjust their demands on a shared system, with little insight into the IT infrastructure installed due to which the power demands of IT loads can exhibit considerable variability throughout the day since they are driven by diverse requirements. Emerging workloads from Graphics Processing Units (GPU) display unique power consumption patterns, frequently leading to significant and nearly instantaneous spikes in power demand. With nearly every watt of electrical energy being processed by power conversion devices multiple times, data centers exhibit a very high density of power electronics, exceeding that of substantial wind and PV installations and nearing the levels found in HVDC converter stations [14] [15]. Consequently, data centers are similarly susceptible to the types of instability and resonance challenges that have recently beset the renewable energy and HVDC sectors in recent years [16], [17].

Considering the substantial and unavoidable expansion of modern data centers which highlights their critical significance; and acknowledging their progressively vital function in the present-day global landscape, it is essential to confront the advancing power demand challenges related to data centers while addressing issues related to their power quality.

C. Approach

Motivated by the above challenges, this paper discusses the technical considerations to help define and refine equipment specifications and sizing for modern data center designs and recommends different power system studies to help overcome these challenges. The structure of this paper is as follows: Section II discusses the technical considerations for designing modern data centers, followed by the technical concerns and challenges discussed above in greater detail. Section III briefly reviews the basic compliance studies for data centers. In addition, it discusses the power system studies suggested for overcoming the challenges discussed in section II and advanced modeling considerations for power system stability analysis. Lastly, section IV presents the conclusions.

II. DESIGN CONSIDERATIONS AND TECHNICAL CONCERNS WITH MODERN DATA CENTERS

A. Design Considerations

General design considerations for data centers include geographical positioning (availability of power, network connectivity, environmental conditions), electrical distribution and cooling infrastructure, IT apparatus (servers, storage solutions, networking equipment), security protocols, redundancy measures, scalability considerations, energy efficiency, financial implications, and compliance with

regulatory standards; all of which necessitate meticulous evaluation and optimization to guarantee dependable data processing and storage functionalities. Once the maximum design IT load capacity is determined after taking into consideration the future expansion, load calculations should consider adding the total mechanical cooling load and any auxiliary load to support the IT infrastructure and to meet the redundancy requirements required by the tier rating classification. When designing the mechanical cooling system, maximizing Power Usage Efficiency (PUE) should be prioritized to improve the overall system efficiency, given the inefficiencies in a data center power system due to multiple power conversions and losses.

Special considerations must be given to network symmetry, load calculations, and load balancing across all three phases to create a balanced system as evenly loaded across all three phases as practical to improve system stability at low frequency to overcome low frequency resonance issues [14] [15]. Also, special attention may need to be given to designing the distribution network to reduce or minimize the zero-sequence impedance of the power system in steady state -for example wye connected loads like Power Supply Units (PSU), to improve overall power system stability [15].

B. Technical Challenges with Modern Data Centers

Modern data centers frequently experience load fluctuations and instantaneous spikes which pose significant challenges to data center power systems. Such rapid changes in power flow, frequency, or voltage can lead to system instability and cause equipment failures unless the power system design is sound and robust from a power system stability perspective. Instantaneous spikes in load fluctuations may require early detection for timely resource and capacity allocation to prevent impacts on system performance or cause equipment degradation and failures.

In addition to the high load variability, the modern data center designers are also challenged with the problem of resonance. This phenomenon in power systems was recently observed in new data centers at both low and high frequencies. Figures 1(a) and 1(b) show one such example, discussed in [14]. Low frequency resonances in Figure 1(a) were at less than the second harmonic frequency, with an oscillation at 11 Hz seen in the amplitude of both the current and voltage measured at the tap box to a server rack [14]. High frequency resonances in Figure 1(b) in the 5-10 kHz range were observed in some other data centers [14]. Such resonance phenomenon may create instability in data center power systems during perturbations which may be caused by frequent load variations and instantaneous spikes, eventually causing equipment failures or system outages.

To overcome the complex challenges discussed above and to analyze the different power quality and power system parameters for modern data centers, different power system studies are recommended as outlined in Section III.



variable frequency drives (VFDs), etc., a power quality and harmonics analysis is recommended to mitigate the harmonics and comply with the IEEE-519 standard. The IEEE-519 standard gives recommended limits for current distortion due to consumer loads and voltage distortion in the utility supply voltage, both referenced at the point of common coupling(PCC). Figure 3 shows an example of a power system one-line diagram depicting the current THD (%) and PF at different branches and voltage THD (%) at different buses. The results from the harmonics analysis can be utilized to include any active or passive filters in the design and avoid any harmonics related issues during the operation of the data center.

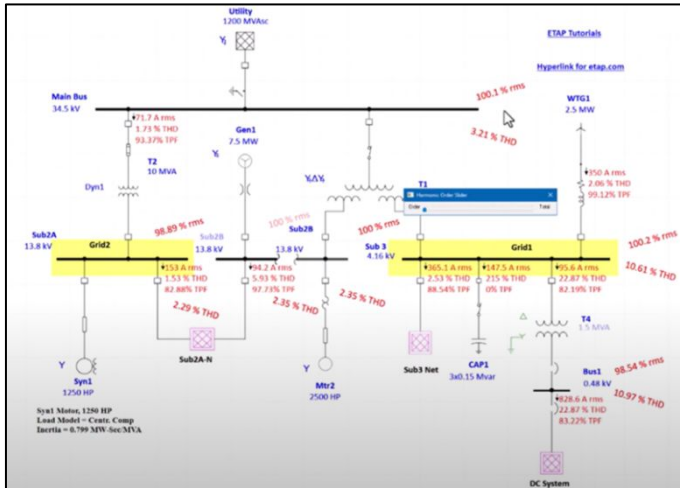


Figure 3 - An Example of a power system SLD depicting the current THD (%) and PF at different branches and voltage THD (%) at different buses

C. Time Domain Load Flow Analysis

A TDLF analysis has diverse applications. It can be used to identify potential points of failure during operations in an existing data center power system, develop and refine equipment specifications, and analyze load, source and system behavior. It can offer a granular and detailed assessment of power quality and power system parameters such as power factor, active and reactive power, active and reactive power losses as a function of time. Additionally, this analysis can help the design engineers analyze and review the annual energy consumption trends, daily power consumption trends, current and voltage waveforms, harmonic distortions etc. at any node of choice. This will facilitate root cause analysis of loading, power factor and voltage issues and develop solutions to overcome these issues. Figure 4 below shows an example of a power system setup for performing a TDLF analysis. Figure 5 shows the plot of total system losses versus time, Figure 6 shows the plot of power factor at the source bus versus time, and Figure 7 depicts the plot of phase-A voltage at a distribution bus versus time, for a given IEEE-34 bus distribution power system test bed.

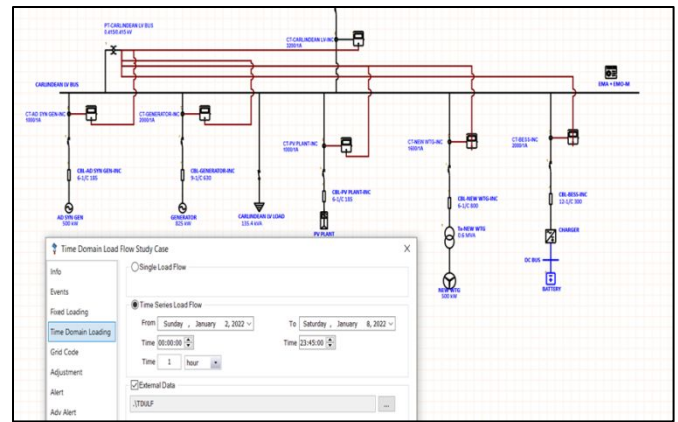


Figure 4 - Example of a partial power system SLD setup for TDLF Analysis.

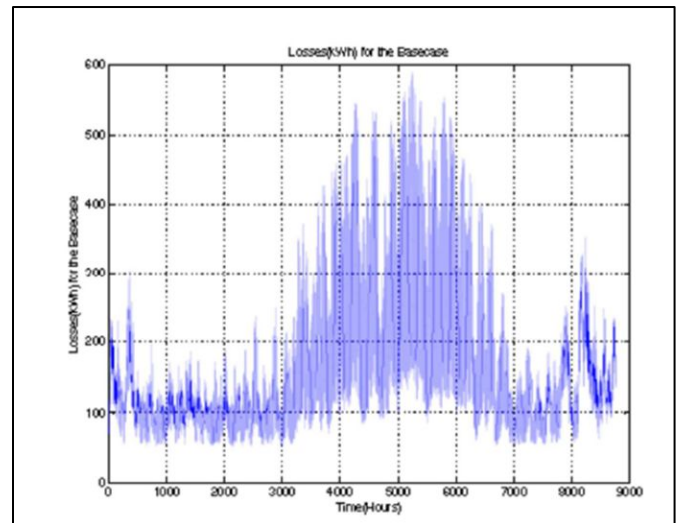


Figure 5 – Plot of total power losses vs time for a given IEEE-34 bus distribution power system test bed.

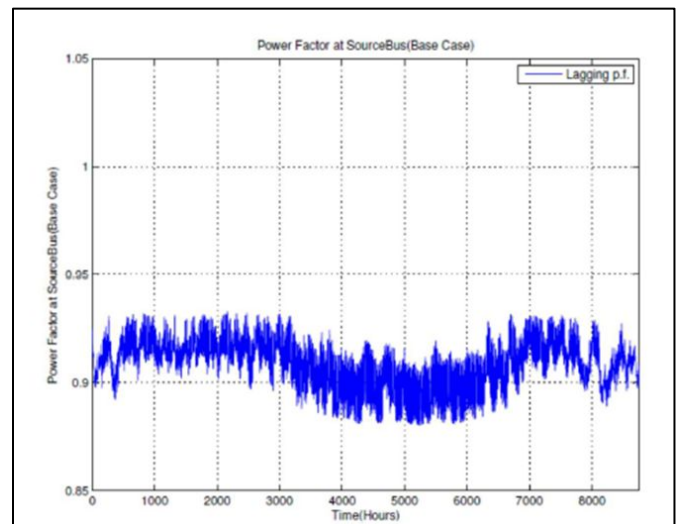


Figure 6 – Plot of power factor at the source bus at the substation vs time for a given IEEE-34 bus distribution power system test bed.

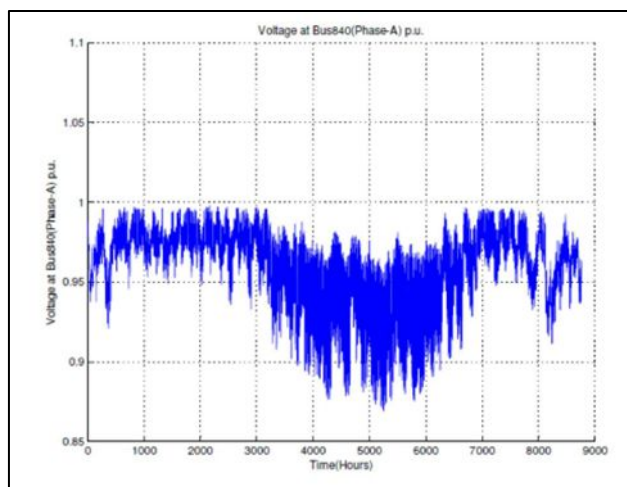


Figure 7 – Plot of phase-A voltage at a load bus vs time for a given IEEE-34 bus distribution power system.

D. Power System Transients Studies

Rapid changes in power flow, frequency, or voltage can lead to system instability, which may potentially cause system blackouts and/or equipment failures. Transient stability analysis can be used to evaluate a power system's dynamic response to perturbations and assess its steady state stability. It can also be used to assess various power system and power quality parameters in data centers during instantaneous spikes in load fluctuations to help develop equipment specifications and improve overall power system design. Advanced impedance modeling of data center equipment for modern data centers may sometimes be necessary for performing system stability analysis [14] [15]. Advanced modeling considerations may include building an impedance model for PSUs, developing building distribution network model by impedance scaling, system modeling, and model reduction based on equivalent source impedance and system stability analysis [15]. The analysis from such studies will help the engineers understand the root cause of system issues like resonance at low and high frequencies, and develop the overall data center power system design and equipment specifications after taking advanced technical considerations and design aspects into account that impact power system stability like network symmetry, load calculations and balancing and minimizing zero sequence impedance of the overall network. Figure 8 shows current and voltage waveforms captured during transient analysis of an off-grid sudden switching-in event of a MV transformer onto a power system in steady state.

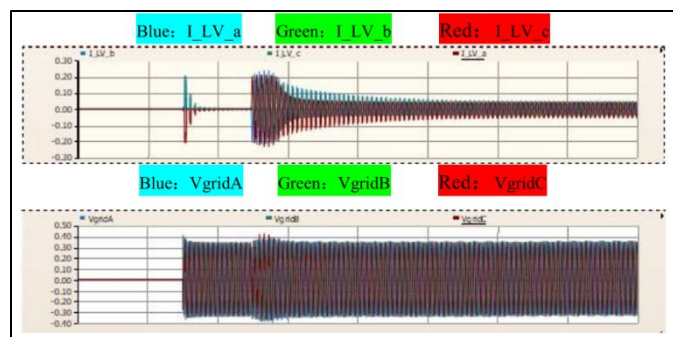


Figure 8 Current and voltage waveforms captured during a transient analysis of an off-grid sudden switching-in event of an MV transformer onto a power system in steady state.

E. System Grounding Study

A system grounding analysis is typically required for complex power systems such as data centers since it affects the system's susceptibility to voltage transients and helps in defining the system protection requirements. The system grounding arrangement is determined by the grounding of the power source. For commercial and industrial systems, the types of power sources generally fall into four broad categories, namely the utility service, generator, transformer, and static power converters. The system grounding on the system fed by the utility service or a customer-owned transformer is determined by the transformer secondary winding configuration. In grid-connected mode, the utility typically serves as the system grounding reference for the power system. Since the loss of utility causes the loss of the system grounding reference, a system grounding reference is required for the data center power system to operate in off-grid mode. For generators, the system grounding is determined by stator winding configuration; for static power converter devices such as rectifiers and inverters, the system grounding is determined by the output stage of the converter.

There are three different types of system grounding, solidly grounded systems, impedance grounded systems (including low and high impedance grounded), and ungrounded systems. Solidly grounded systems have no intentional impedances to ground to limit fault currents, thus allowing higher fault current magnitudes, facilitating protection operation and system device coordination. Impedance-grounded systems allow control of ground fault currents by introducing an intentional impedance in the ground fault current path. An ungrounded system is a system where there is no intentional connection of the system to ground.

A system grounding analysis can be performed in ETAP or a similar software if the impedances of the source and the system are known. This study is used to establish a power system's grounding performance. The Coefficient of Ground (COG) is defined below per IEEE C62.92.1-6 standard:

$$\text{Coefficient of Grounding (COG)} = \frac{\text{Highest Unfaulted Phase} - \text{to} - \text{Ground Voltage During a Ground Fault}}{\text{Phase} - \text{to} - \text{Phase Voltage with Ground Fault Removed}}$$

Figure 9 Coefficient of Grounding formula as defined by IEEE C62.91.1-6 standards.

Effective grounding of power systems is characterized by a $\text{COG} < 0.8$. A COG lower than 0.8 is recommended to limit the transient over voltages (TOVs) within the power system and can be achieved through proper system grounding design. Lowering the COG results in lower TOVs, leading to an improved overall system grounding performance. Figure 10 shows an example of a system grounding analysis performed for an off-grid power system supported by two generators in parallel and renewable sources offline. A single-line-to-ground (SLG) fault occurs on phase A of the 13.2 kV medium voltage(MV) switchgear bus, leading to temporary over-voltages(TOVs) on phases B and C of the switchgear bus. The MV switchgear is grounded using a zig-zag transformer through a neutral grounding resistor(NGR). Since the MV switchgear being grounded through an NGR, the resulting TOVs across the line-neutral voltages on phases B and C due phase-A ground fault are limited to 10.18kV and 12.75kV respectively, compared to 13.2 kV line-neutral voltages that would have otherwise developed across phases B and C on an ungrounded power system.

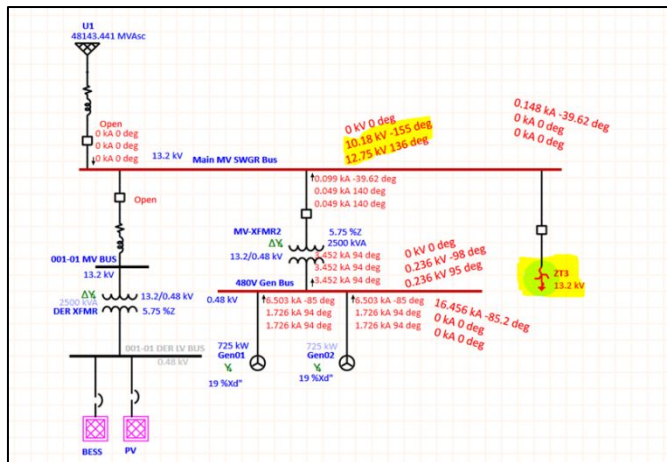


Figure 10 System Grounding Analysis model Example – Transient over voltages observed on B & C Phases B & C of a 13.2kV MV Switchgear Bus grounded using a zig-zag transformer through an NGR during an SLG fault on A phase for an off-grid power system islanded on paralleled generators.

IV. CONCLUSIONS

Globally data centers are growing rapidly as a result of exponential growth in cloud computing, AI, IoT, as well as the intensifying demand for high-bandwidth applications. The energy demand of the data centers is becoming a significant share of the total global energy demand, requiring special attention of the design engineers during the early design phases to address the issues that were historically not of significant concern, but are concerns with modern data centers.

This paper discussed various power system studies recommended for designing modern data center power systems in addition to general design considerations. In addition, special considerations for modern data center power systems were discussed, which may necessitate advanced modeling and power system analysis to design data center power systems and develop equipment specifications.

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